MILITARY SPECIFICATION SHEET

RUBBER, ETHYLENE-PROPYLENE, HYDRAZINE RESISTANT, O-RINGS SIZES AND TOLERANCES

This specification sheet is approved by the Air Force Materials Laboratory, MXA, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the o-rings described herein shall consist of this document and the latest issue of specification MIL-R-83412(USAF).

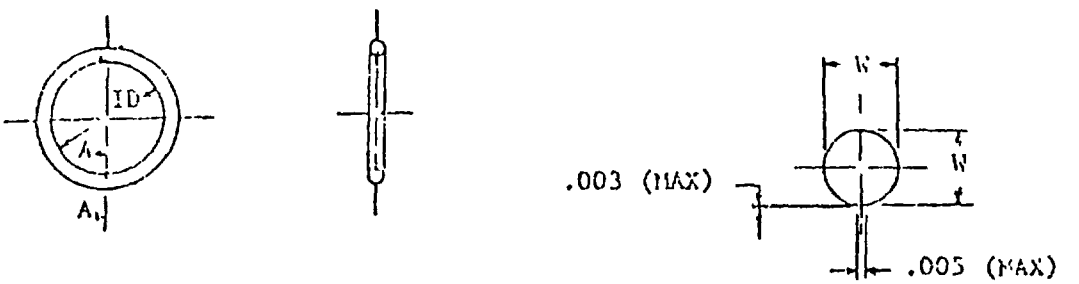


FIGURE 1. O-ring, front, side, and cross-section.

Purpose. The purpose of this specification sheet is to set up a standardized part numbering system for o-rings procured to MIL-R-83412(USAF), Type I.

Intended use. These o-rings are recommended for use in packing glands such as specified in MIL-G-5514 (See Note).

Part numbers. When this document is used for procurement of o-rings, part numbers shall consist of the following:

1. The letter M and the specification sheet number.

2. A dash followed by the appropriate dash number for Table I or Table II.

Example 1	Example 2
M83412/1-214	M83412/1-340

NOTE: The following sizes are not included in MIL-G-5514: 029 to 050, 102 to 109, 150 to 178, 201 to 209, 248 to 284, 309 to 324, 350 to 395, 461 to 475, and 901 to 932.

FSC 5330

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TABLE I. O-ring sizes

DASH NO.	ACTUAL SIZE W		ID	
001	.040	±.003	.029	±.004
002	.050		.042	
003	.060		.056	
004	.070	±.003	.070	±.005
005			.101	
006			.114	
007			.145	
008			.176	
009			.208	
010			.239	
011			.301	
012			.364	
013			.426	
014			.489	
015			.551	±.007
016			.614	±.009
017			.676	
018			.739	
019			.801	
020			.864	
021			.926	
022			.989	±.010
023			1.051	
024			1.114	
025			1.176	±.011
026			1.239	
027			1.301	
028			1.364	±.013
029			1.489	
030			1.614	
031			1.739	
032			1.864	
033			1.989	±.018
034			2.114	
035			2.239	
036			2.364	
037			2.489	
038			2.614	±.020
039			2.739	
040			2.864	

DASH NO.	ACTUAL SIZE W		ID	
041	.070	±.003	2.989	±.024
042			3.239	
043			3.489	
044			3.739	±.027
045			3.989	
046			4.239	±.030
047			4.489	
048			4.739	
049			4.989	±.037
050			5.239	
102	.103	±.003	.049	±.005
103			.081	
104			.112	
105			.143	
106			.174	
107			.206	
108			.237	
109			.299	
110			.362	
111			.424	
112			.487	
113			.549	±.007
114			.612	±.009
115			.674	
116			.737	
117			.799	±.010
118			.862	
119			.924	
120			.987	
121			1.049	
122			1.112	
123			1.174	±.012
124			1.237	
125			1.299	
126			1.362	
127			1.424	
128			1.487	
129			1.549	±.015
130			1.612	

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TABLE I. O-ring sizes. - Continued

DASH NO.	ACTUAL SIZE W		ID	
131	.103	±.003	1.674	±.015
132			1.737	
133			1.799	
134			1.862	
135			1.925	±.017
136			1.987	
137			2.050	
138			2.112	
139			2.175	
140			2.237	
141			2.300	±.020
142			2.362	
143			2.425	
144			2.487	
145			2.550	
146			2.612	
147			2.675	±.022
148			2.737	
149			2.800	
150			2.862	
151			2.987	±.024
152			3.237	
153			3.487	
154			3.737	±.028
155			3.987	
156			4.237	±.030
157			4.487	
158			4.737	
159			4.987	±.035
160			5.237	
161			5.487	
162			5.737	
163			5.987	
164			6.237	±.040
165			6.487	
166			6.737	
167			6.987	
168			7.237	±.045
169			7.487	
170			7.737	

DASH NO.	ACTUAL SIZE W		ID	
171	.103	±.003	7.987	±.
172			8.237	±.
173			8.487	
174			8.737	
175			8.987	
176			9.237	±.
177			9.487	
178			9.737	
201	.139	±.004	.171	±.
202			.234	
203			.296	
204			.359	
205			.421	
206			.484	
207			.546	±.
208			.609	±.
209			.671	
210			.734	±.
211			.796	
212			.859	
213			.921	
214			.984	
215			1.046	
216			1.109	±.0
217			1.171	
218			1.234	
219			1.296	
220			1.359	
221			1.421	
222			1.484	±.
223			1.609	
224			1.734	
225	.139	±.004	1.859	±.
226			1.984	
227			2.109	
228			2.234	±.
229			2.359	
230			2.484	
231			2.609	

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TABLE I. O-ring sizes. - Continued

DASH NO.	ACTUAL SIZE W		ID	
232	.139	±.004	2.734	±.024
233			2.859	
234			2.984	
235			3.109	
236			3.234	
237			3.359	
238			3.484	
239			3.609	±.028
240			3.734	
241			3.859	
242			3.984	
243			4.109	
244			4.234	±.030
245			4.359	
246			4.484	
247			4.609	
248			4.734	
249			4.859	±.035
250			4.984	
251			5.109	
252			5.234	
253			5.359	
254			5.484	
255			5.609	
256			5.734	
257			5.859	
258			5.984	
259			6.234	±.040
260			6.484	
261			6.734	
262			6.984	
263			7.234	±.045
264			7.484	
265			7.734	
266			7.984	
267			8.234	±.050
268			8.484	
269			8.734	
270			8.984	
271			9.234	±.055
272			9.484	

DASH NO.	ACTUAL SIZE W		ID	
273	.139	±.004	9.734	±.055
274			9.984	
275			10.484	
278			10.984	±.065
277			11.484	
278			11.984	
279			12.984	
280			13.984	
281			14.984	
282			15.955	±.075
283			16.955	±.080
284			17.955	±.085
309	.210	±.005	.412	±.005
310			.475	
311			.537	±.007
312			.600	±.009
313			.662	
314			.725	±.010
315			.787	
316			.850	
317			.912	
318			.975	±.010
319			1.037	
320			1.100	±.012
321			1.162	
322			1.225	
323			1.287	
324			1.350	
325			1.475	±.015
326			1.600	
327			1.725	
328			1.850	
329			1.975	±.018
330			2.100	
331			2.225	
332			2.350	
333			2.475	±.020
334			2.600	
335			2.725	
336			2.850	

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TABLE I. C-ring sizes. - Continued

DASH NO.	ACTUAL SIZE W		ID	
337	.210	±.005	2.975	±.024
338			3.100	
339			3.225	
340			3.350	
341			3.475	
342			3.600	±.028
343			3.725	
344			3.850	
345			3.975	
346			4.100	
347			4.225	±.030
348			4.350	
349			4.475	
350			4.600	
351			4.725	
352			4.850	
353			4.975	±.037
354			5.100	
355			5.225	
356			5.350	
357			5.475	
358			5.600	
359			5.725	
360			5.850	
361			5.975	
362			6.225	±.040
363			6.475	
364			6.725	
365			6.975	
366			7.225	±.045
367			7.475	
368			7.725	
369			7.975	
370			8.225	±.050
371			8.475	
372			8.725	
373			8.975	
374			9.225	±.055
375			9.475	
376			9.725	
377			9.975	

DASH NO.	ACTUAL SIZE W		ID	
378	.210	±.005	10.475	±.060
379			10.975	
380			11.475	±.065
381			11.975	
382			12.975	
383			13.975	±.070
384			14.975	
385			15.955	±.075
386			16.955	±.080
387			17.955	±.085
388			18.955	±.090
389			19.955	±.095
390			20.955	
391			21.955	±.100
392			22.940	±.105
393			23.940	±.110
394			24.940	±.115
395			25.940	±.120
425	.275	±.006	4.475	±.033
426			4.600	
427			4.725	
428			4.850	
429			4.975	±.037
430			5.100	
431			5.225	
432			5.350	
433			5.475	
434			5.600	
435			5.725	
436			5.850	
437			5.975	
438			6.225	±.040
439			6.475	
440			6.725	
441			6.975	
442			7.225	±.045
443			7.475	
444			7.725	
445			7.975	
446			8.475	±.055

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TABLE I. O-ring sizes. - Continued

DASH NO.	ACTUAL SIZE			
	W		ID	
447	.275	±.006	8.975	±.055
448	↓	↓	9.475	↓
449	↓	↓	9.975	↓
450	↓	↓	10.475	±.060
451	↓	↓	10.975	↓
452	↓	↓	11.475	↓
453	↓	↓	11.975	↓
454	↓	↓	12.475	↓
455	↓	↓	12.975	↓
456	↓	↓	13.475	±.070
457	↓	↓	13.975	↓
458	↓	↓	14.475	↓
459	↓	↓	14.975	↓
460	↓	↓	15.475	↓
461	↓	↓	15.955	±.075
462	↓	↓	16.455	↓
463	↓	↓	16.955	±.080
464	↓	↓	17.455	±.085
465	↓	↓	17.955	↓
466	↓	↓	18.455	↓
467	↓	↓	18.955	±.090
468	↓	↓	19.455	↓
469	↓	↓	19.955	±.095
470	↓	↓	20.955	↓
471	↓	↓	21.955	±.100
472	↓	↓	22.940	±.105
473	↓	↓	23.940	±.110
474	↓	↓	24.940	±.115
475	↓	↓	25.940	±.120

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TABLE II. Straight thread tube fitting boss gaskets.

DASH NO.	TUBE OUTSIDE DIAMETER	W CROSS-SECTION		I.D.	
901	094	.056	±.003	.185	±.005
902	.125	.064		.239	
903	.188	↓		.301	
904	.250	.072		351	
905	.312	↓		.414	
906	.375	.078		.468	↓
907	.438	.082		.530	±.007
908	.500	.087		.5 4	±.009
909	.562	.097	↓	.70	↓
910	.625	↓		.755	
911	.688	.116	±.004	.863	↓
912	.750	↓	↓	.924	
913	.812			.986	±.010
914	.875			1.047	↓
916	1.000	↓		1.171	↓
918	1.125			1.355	±.012
920	1.250	.118		1 475	±.014
924	1.500	↓		1 720	↓
928	1.750			1.690	±.018
932	2.000	↓	↓	2 337	↓

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NOTICE 2
10 December 1992

MILITARY SPECIFICATION

RUBBER, ETHYLENE-PROPYLENE, HYDRAZINE RESISTANT, O-RINGS SIZES AND TOLERANCES

MIL-R-83412/1(USAF), dated 17 July 1979, has been reviewed and determined to be valid for use in acquisition.

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